Frequentist and Bayesian Analysis of the Matter–Antimatter Asymmetry in $B^{\pm} \to K^{\pm}K^{+}K^{-}$ Decays at LHCb

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We measure the raw charge asymmetry $A_{\rm raw} = [N(B^-) - N(B^+)]/[N(B^-) + N(B^+)]$ of $B^{\pm} \to K^{\pm}K^{+}K^{-}$ decays in the 2011 LHCb Open Data record with an extended unbinned maximum-likelihood fit and with a Bayesian analysis of the same likelihood. On 19,731 candidates we find $A_{\rm raw} = -0.0441 \pm 0.0085\,(\text{stat}) \pm 0.0072\,(\text{syst})$ (Frequentist) and $-0.0443 \pm 0.0085 \pm 0.0070$ (Bayesian), 5.2 standard deviations from zero. The methods agree in value, uncertainty and coverage, as verified on toy-MC experiments, and the dominant systematic uncertainty is the particle-identification requirement. In the Dalitz region of the published analysis we find -0.234 ± 0.025, reproducing -0.239 ± 0.020. $A_{\rm raw}$ is not corrected for production and detection asymmetries and is not a measurement of A_{CP}.

I. INTRODUCTION

The observable of this work is the raw charge asymmetry of the signal yields of $B^{\pm} \to K^{\pm}K^{+}K^{-}$ decays,

$$A_{\rm raw} = \frac{N(B^-) - N(B^+)}{N(B^-) + N(B^+)}, \tag{1}$$

which in three-body charmless B decays can be studied globally and locally over the Dalitz plane [1, 2]. It differs from the CP asymmetry A_{CP} by the production asymmetry A_P of $B^{\pm}$ in pp collisions and by the detection asymmetry A_D of the final state [1],

$$A_{CP} \approx A_{\rm raw} - A_P - A_D. \tag{2}$$

Neither correction is applied here, so $A_{\rm raw}$ is not a measurement of A_{CP}; the corrections of the published analysis are discussed in Sec. IX.

We use the LHCb Open Data record for this decay [3, 4], a subset of the 2011 data on which the first evidence of an inclusive CP asymmetry was published [5], to ask: *to what extent do Frequentist likelihood fits and Bayesian inference produce consistent, well-controlled estimates of* $A_{\rm raw}$*?* The aim is a methodological comparison of the two frameworks on the same data and on the same likelihood, so that any difference comes from the inference and not from the model; the record lacks the control channel needed for A_P and A_D. Secs. II–III describe the data and the common model, Secs. IV–V the two fits, and Secs. VI–IX the systematic study, the validation, the Dalitz plane and the comparison, which includes a machine-learning selection. The main results are in Table I.

II. DATA

*In short: the record, the common preselection (*23,895 *candidates) and the fit window (*19,731 *candidates); the spectrum peaks at* 5284 *MeV for both charges and both polarities, and the counting asymmetry is* -0.043 ± 0.007.

A. Dataset, Preselection and Mass

The dataset is the CERN Open Data record 4900 [3]: $B^{\pm} \to h^{\pm}h^{+}h^{-}$ candidates collected by LHCb in 2011 at $\sqrt{s} = 7\,\text{TeV}$, already through trigger and offline selection, split by magnet polarity (8,556,118 candidates), and a phase-space simulation of $B^{\pm} \to K^{\pm}K^{+}K^{-}$ (50,000 decays, no asymmetry) used only to test the local-asymmetry procedure (Sec. VIII). The same preselection is used in all analyses,

$$\texttt{H*_ProbPi} < 0.5 \ \wedge\ \texttt{H*_ProbK} > 0.5 \ \wedge\ \neg\texttt{H*_isMuon},$$

for each of the three tracks, and retains 23,895 candidates (0.28%). $H1$ is always the kaon of charge opposite to the B, so the Dalitz variables are $m^2(H1, H2)$ and $m^2(H1, H3)$.

The three tracks are assigned the charged-kaon mass and their four-momenta are summed to reconstruct $m(KKK)$. The nominal fit window, common to the two analyses, is $[5100, 5600]$ MeV and contains 19,731 candidates (11,636 magnet down, 8095 magnet up). Below ∼5150 MeV the background rises, which a single exponential cannot describe; the window includes the start of this rise, described by a second exponential in the background model (Secs. IV–V).

B. Data Exploration

Figure I shows the spectrum in the fit window. The peak at $m \approx 5284\,\text{MeV}$ over a falling combinatorial background has the same shape for the two charges and the two polarities, which allows a mass and a resolution common to B^+ and B^- in the model (Sec. III). The

Method	Sample	A_{raw}	Statistical	Systematic	Section
Counting (no fit)	fit window, 19,731	-0.0432	0.0071	–	II B
Frequentist fit	fit window, 19,731	-0.0441	0.0085	0.0072	IV, VI
Bayesian fit	fit window, 19,731	-0.0443	0.0085	0.0070	V, VI
Bayesian fit, D^0 veto	18,744	-0.0496	0.0088	–	–
Bayesian fit, ML selection	score > 0.5, 15,709	-0.0451	0.0087	–	IX
Frequentist fit, ML selection	score > 0.5, 15,709	-0.0447	0.0086	–	IX
Bayesian fit, Dalitz region of Ref. [5]	2146	-0.234	0.025	–	VIII
Published analysis (same data) [5]	different selection and model	-0.060	0.007	–	–

TABLE I. Summary of the raw asymmetry A_{raw} (not corrected for production and detection asymmetries). The Bayesian value is the posterior median and its standard deviation; the systematic uncertainty is given only for the nominal fits (Sec. VI). The published A_{raw} comes from a different selection and a model with additional background components, so it is not directly comparable with the first rows.

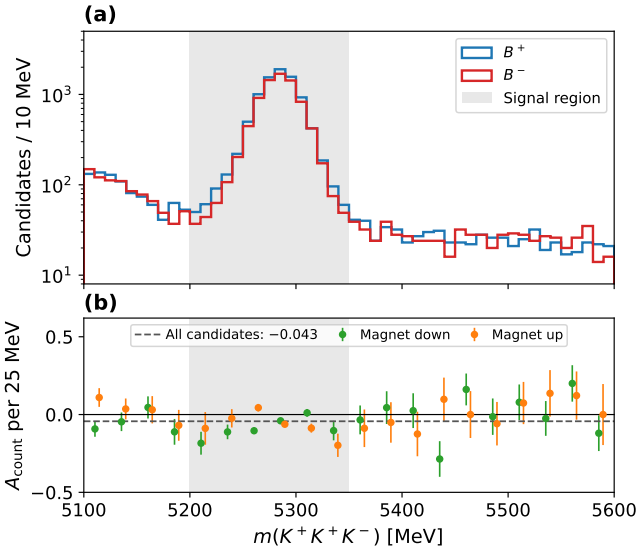


FIG. I. (a) Reconstructed $B \to KKK$ invariant mass in the nominal fit window $[5100, 5600]$ MeV for B^+ and B^-, logarithmic scale; the signal region $[5200, 5350]$ MeV is shaded. (b) Counting asymmetry in 25 MeV bins of the mass for the two magnet polarities, and for all candidates (dashed); the two polarities agree within their uncertainties and the asymmetry is similar in the peak and in the sidebands.

counting asymmetry of all candidates, signal and background together, is $A_{\text{count}} = (N^- - N^+)/(N^- + N^+) = -0.043 \pm 0.007$ (binomial uncertainty); the fits of Secs. IV and V refine it by separating signal and background.

C. Additional Results

Branches and cut flow. The analysis reads momentum, particle-identification likelihoods, muon flag and charge of each track; the kaon-identification requirement on the three tracks is by far the most selective.

Charge combinations. $H1$ is always the kaon of charge opposite to the B and $H2$, $H3$ the same-sign pair, so the Dalitz variables are the two opposite-sign invariant masses.

Shared tracks and window. Only 42 candidates of the fit window (0.2%) share a track with another candidate and all are kept; the mass is computed with the kaon hypothesis on all tracks, and the window includes the start of the rise of the background below ~ 5150 MeV.

III. MASS MODEL AND LIKELIHOOD

In short: one extended likelihood shared by the two analyses: a Gaussian signal with common mean and width, a double-exponential background for each charge and A as a parameter. The mass spectrum of each charge sample is described by a signal-plus-background mixture on the nominal fit window $[5100, 5600]$ MeV (Sec. II A),

$$f^{\pm}(m) = N_s^{\pm}\, S(m; \mu, \sigma) + N_b^{\pm}\, B^{\pm}(m), \qquad (3)$$

where S is a Gaussian truncated to the window, with mean μ and width σ shared by the two charges, and the background is a double exponential with a shape of its own for each charge,

$$B^{\pm}(m) = f_{\pm}\, E(m; \lambda_1^{\pm}) + (1 - f_{\pm})\, E(m; \lambda_2^{\pm}),$$

with $E(m; \lambda) \propto e^{-\lambda m}$ a truncated exponential normalized on the window. The steep component (λ_1) describes the rise of the background towards the lower edge of the window, the slow one (λ_2) the combinatorial background under and above the peak. The ordering $\lambda_1 > \lambda_2$ removes the label-switching symmetry of the mixture. Signal and background yields are split between the charges as

$$N_s^{\pm} = \tfrac{N_s}{2}(1 \mp A), \qquad N_b^{\pm} = \tfrac{N_b}{2}(1 \mp A_b),$$

so that $A = (N_s^- - N_s^+)/(N_s^- + N_s^+)$, the raw asymmetry A_{raw} of Eq. (1), is itself a parameter of the fit, the parameter of interest (POI, all the others being nuisance parameters), and the background asymmetry A_b is free. The parameter vector has twelve components, $\theta = (N_s, N_b, A, A_b, \mu, \sigma, \lambda_1^{\pm}, \lambda_2^{\pm}, f_{\pm})$.

Given the data $D = \{(m_i, q_i)\}_{i=1}^{N}$ (mass and charge of each candidate), the extended unbinned likelihood of the two charge samples is

$$\mathcal{L}(D \mid \theta) \propto e^{-(N_s+N_b)} \prod_{i:\, q_i=+1} \Big[N_s^+ S(m_i) + N_b^+ B^+(m_i)\Big] \times \prod_{i:\, q_i=-1} \Big[N_s^- S(m_i) + N_b^- B^-(m_i)\Big], \quad (4)$$

where $S(m_i) \equiv S(m_i; \mu, \sigma)$. The likelihood is extended (the yields are parameters, hence the Poisson term); it is the common ground of the two mass analyses: Sec. IV maximizes it and Sec. V multiplies it by a prior. Since both use this model unchanged, any difference between their results can only come from the inference.

A. Additional Results

Minimal model. A six-parameter model on a narrower window gives $A_{\text{raw}} = -0.0497$ (68% CI $[-0.058, -0.041]$), compatible with the nominal result within its uncertainty; it is the model of the low-statistics comparison.

IV. FREQUENTIST ANALYSIS

In short: the likelihood maximized for the two charges together gives $N_s = 15{,}328 \pm 133$ *and* $\hat{A} = -0.0441$ *with Minos intervals;* $A = 0$ *is rejected at* 5.2σ *and the interval has nominal coverage in* 1000 *bootstrap replicas.* The Frequentist analysis estimates A_{raw} by maximizing the extended unbinned likelihood of Eq. (4) with `iminuit` [6]; the uncertainty and the test of $A = 0$ come from the profile likelihood, the other parameters being nuisance parameters (Secs. IV A–IV B).

A. Mass Model and Fit Strategy

The common mass model of Sec. III (Eq. (3)) was fitted on $[5100, 5600]$ MeV: a Gaussian signal with mean μ and width σ shared by the two charges and a double-exponential background with shape and yield of its own for each charge, twelve parameters as in the Bayesian fit, through `iminuit.cost.ExtendedUnbinnedNLL`; the minimization started from an unbinned fit of the sidebands (the window without $[5200, 5350]$ MeV). The asymmetry was estimated in two ways: in the *simultaneous* fit the two charges share $N_s^{\pm} = N_s(1 \mp A)/2$ and A is a parameter of the likelihood; in the *separate* fits each charge was fitted alone (seven free parameters each) and $A_{\text{raw}} = (N_s^- - N_s^+)/(N_s^- + N_s^+)$ was computed from the two yields. They give -0.0441 ± 0.0085 and -0.0442 ± 0.0087: sharing μ and σ between a particle and its antiparticle is adequate, and the simultaneous fit, which gives the profile-likelihood interval directly, is the nominal one. It gives $N_s = 15{,}328 \pm 133$ and $N_b = 4{,}403 \pm 82$, against the Bayesian posterior medians $15{,}328 \pm 133$ and 4404 ± 82 (Sec. V C). The shapes were not tuned on this fit, and the alternatives are systematic variants (Sec. VI). The nominal fit has $\chi^2/\text{ndof} = 160/88$ ($p = 4.8 \cdot 10^{-6}$) in 2×50 bins of 10 MeV; a single exponential is rejected (294/92, $p = 5.7 \cdot 10^{-23}$) because the background rises towards the lower edge of the window (Sec. II A). The fit of the two charges is shown in Fig. II.

B. Profile Likelihood and Confidence Interval

The maximum-likelihood estimate is $\hat{A} = -0.0441$ (Hesse uncertainty 0.0085), with `MINOS` [7] intervals 68% CI $[-0.0526, -0.0356]$ and 95% CI $[-0.0607, -0.0275]$; the uncertainties are statistical only (the systematic one is in Sec. VI). The profile likelihood ratio $q(A) = -2\log[L(A, \hat{\hat{\theta}})/L(\hat{A}, \hat{\theta})]$, with the other eleven parameters re-minimized at each A (`mnprofile`), is shown in Fig. III; by Wilks' theorem [8] it follows a χ^2 distribution with one degree of freedom at the true value of A, so the intervals are the sets $q \le 1$ and $q \le 3.84$. For the null hypothesis, $q(0) \simeq 27$, $p = 2.05 \times 10^{-7}$, 5.20σ: no asymmetry is rejected, the counterpart of the Savage-Dickey test of Sec. V D ($A = 0$ at 5.2 posterior standard deviations); the p-value is not the probability of H_0 and the Bayes factor depends on the prior width, but the conclusion is the same. The result agrees with the Bayesian one (Sec. V C: -0.0443, 68% CI $[-0.0528, -0.0358]$), as expected with flat priors and $\sim 2 \times 10^4$ events.

C. Coverage of the Wilks Intervals

Wilks' theorem is asymptotic, so the coverage was checked on the data with 1000 bootstrap replicas of the 19,731 candidates, refitting the nominal model on each (non-converged fits discarded); with $A_{\text{true}} = -0.0441$, the 68% interval of a replica contains it if and only if $q(A_{\text{true}}) \le 1$. The fraction is $68.4 \pm 1.5\%$ against 68.3% nominal (Fig. IV). It agrees with the toy-MC check of the Bayesian interval ($65.0 \pm 3.3\%$, 200 experiments); the two checks are not equivalent, since the Bayesian pseudo-datasets come from the fitted model (a closure test) while the replicas are resampled from the data and carry the features that the model does not describe, such as the signal tails.

D. Additional Results

Neyman construction and momentum. Inverting the test of A_{true} with toy-MC experiments gives the same interval as the likelihood ratio, and A_{raw} does not depend on the momentum of the B.

Uncertainties of the twelve parameters. The standard deviations from the Hesse matrix, the bootstrap of the

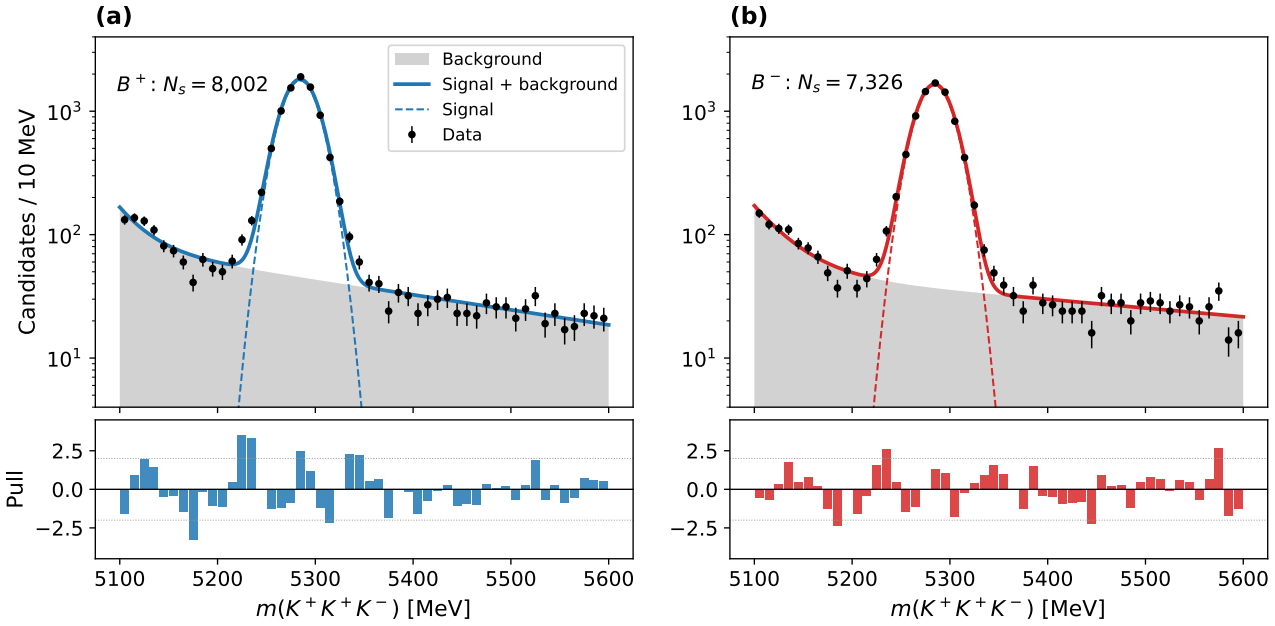


FIG. II. Simultaneous fit of (a) B^+ and (b) B^- in 10 MeV bins, logarithmic scale: data, signal (dashed), double-exponential background (grey) and their sum; below, the pulls $(n_i - \mu_i)/\sqrt{\mu_i}$ of the fit, with the ± 2 lines. The fitted signal yields are 8002 and 7326.

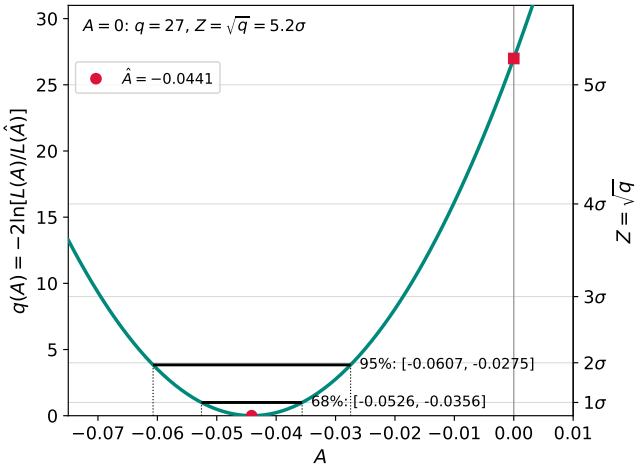


FIG. III. Profile likelihood ratio $q(A)$ of the simultaneous fit, with the other eleven parameters re-minimized at each A (right axis: $Z = \sqrt{q}$). The horizontal segments are the 68% and 95% Minos intervals ($q \leq 1$, $q \leq 3.84$); at $A = 0$, $q \simeq 27$, i.e. 5.2σ.

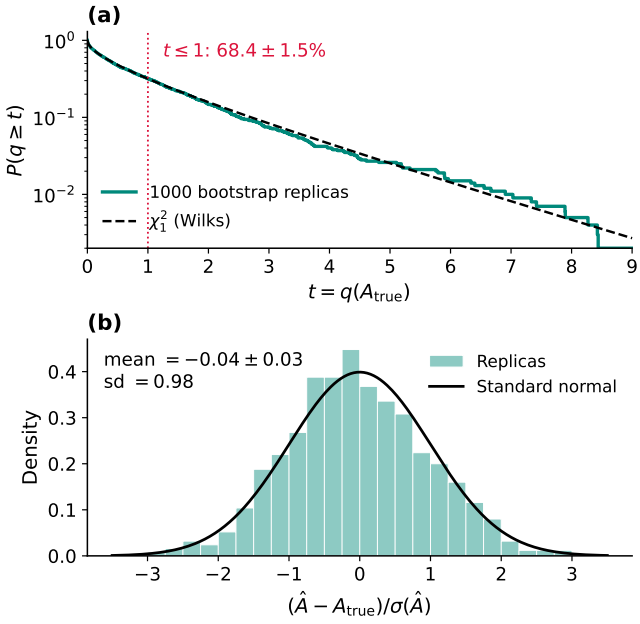


FIG. IV. Coverage check with 1000 bootstrap replicas of the data. (a) Survival function of $t = q(A_{\text{true}})$ against the χ^2_1 distribution expected from Wilks' theorem; the dotted line is the 68.3% limit ($t = 1$). (b) Pulls $(\hat{A} - A_{\text{true}})/\sigma(\hat{A})$ of the replicas against a standard normal.

data and the posterior agree for A, A_b, μ and σ; the bootstrap differs for the two yields, because it keeps the total number of candidates fixed, and for the weakly constrained steep slope λ_1^+.

Starting values. 29 of 30 fits from random starting points reach the same minimum.

V. BAYESIAN ANALYSIS

In short: the same likelihood with uniform priors, sampled with MCMC: $A_{\rm raw} = -0.0443$ (68% interval $[-0.0528, -0.0358]$) with converged chains; the Bayes factor of $A = 0$, $BF_{10} \approx 9 \times 10^3$, depends on the prior width. The Bayesian analysis takes the likelihood of Sec. III as it is and adds only a prior and a sampler; it changes the question asked of the data, not the model.

A. Priors and Posterior

Bayes' theorem combines the likelihood of Eq. (4) (Sec. III) with a prior $\pi(\theta)$ into the posterior

$$p(\theta \mid D) = \frac{\mathcal{L}(D \mid \theta)\,\pi(\theta)}{p(D)}, \tag{5}$$

The priors are uniform, with independent ranges except for the ordering of the background slopes: $N_s, N_b \in (0, 5N)$ with $N = 19{,}731$, $A, A_b \in (-1, 1)$, $\mu \in (5260, 5310)$ MeV, $\sigma \in (3, 40)$ MeV, $\lambda_1^{\pm} \in (10^{-5}, 0.1)$ MeV^{-1}, $\lambda_2^{\pm} \in (10^{-5}, \lambda_1^{\pm})$, $f_{\pm} \in (0, 1)$. The prior density is therefore a constant on the allowed region Θ,

$$\pi(\theta) \propto \mathbf{1}\,[\theta \in \Theta]\,, \tag{6}$$

and inside it the posterior is proportional to the likelihood. With $\sim 2 \times 10^4$ events the posterior is dominated by the likelihood, as a prior-sensitivity check confirms for A. The evidence $p(D)$ is a constant: the MCMC samples $\mathcal{L}\,\pi$ without it, and the Savage-Dickey ratio of Sec. V D avoids it as well; the marginal posterior of A is the histogram of the A component of the chain.

B. MCMC Sampling and Convergence Diagnostics

The posterior was sampled with the affine-invariant ensemble sampler of Goodman and Weare [9] as implemented in `emcee` [10]: 32 walkers started around the maximum of the posterior, 25,000 steps with the first 1/6 discarded as burn-in (6.7×10^5 samples, acceptance 0.38). The integrated autocorrelation time is $\tau \in [147, 183]$ steps, so the chain is $N/\tau \geq 113$ autocorrelation times long for every parameter, above the threshold of 50 [10] ($\gtrsim 3600$ effectively independent samples). Two runs started uniformly over the whole prior box converge to the same single mode ($A_{\rm raw} = -0.0446$ and -0.0442).

C. Posterior Results and Credible Intervals

The nominal model with the posterior medians coincides with the fit of Fig. II: it follows the spectrum over two orders of magnitude, and the pulls show signal tails heavier than a Gaussian, whose effect on $A_{\rm raw}$ is negligible. The marginal posterior on the raw asymmetry gives (median and equal-tailed credible intervals, CI)

$$\begin{gathered} A_{\rm raw} = -0.0443, \\ 68\%\ \text{CI}\ [-0.0528, -0.0358], \\ 95\%\ \text{CI}\ [-0.0608, -0.0276]. \end{gathered}$$

The posterior is very close to Gaussian (standard deviation 0.0085), so equal-tailed and highest-posterior-density intervals coincide; with uniform priors its mode, -0.0441, is the maximum-likelihood estimate and the result can be compared directly with the Frequentist fit (Sec. IV). The other physics parameters are $N_s = 15{,}328 \pm 133$, $N_b = 4404 \pm 82$, $A_b = -0.040 \pm 0.018$, $\mu = 5284.39 \pm 0.16$ MeV and $\sigma = 18.00 \pm 0.14$ MeV. A is almost uncorrelated with them, which is why profiling (Frequentist) and marginalizing (Bayesian) the nuisance parameters give the same result; the largest correlation, -0.17, is with A_b: conditioning on $A_b = 0$ moves A to -0.0475, as the fit with a charge-symmetric background does (-0.0471, Sec. VI).

D. Bayes Factors

The Bayes factor compares two hypotheses through their evidences, $BF_{10} = P(D \mid H_1)/P(D \mid H_0)$, and requires proper priors. For the nested hypothesis H_0: $A = 0$ against H_1: A free, with shared priors on the other parameters, it reduces to the Savage-Dickey ratio [11],

$$BF_{01} = \frac{p(A = 0 \mid D)}{\pi(A = 0)}, \qquad \pi(A = 0) = \tfrac{1}{2}.$$

$A = 0$ lies 5.2 posterior standard deviations from the median, where the chain has essentially no samples, so $p(A = 0 \mid D)$ was taken from the Gaussian approximation of the marginal posterior: $BF_{01} \approx 1.1 \times 10^{-4}$ ($BF_{10} \approx 9 \times 10^3$; Fig. Va), decisive evidence against $A = 0$ on Jeffreys' scale [12]. The value scales with the prior width (Ockham factor): with $A \in (-0.3, 0.3)$, BF_{01} drops by a factor 3.3 and the conclusion is unchanged.

The same test on the background asymmetry shows the Ockham factor at work: $A_b = -0.040 \pm 0.018$ is 2.3 standard deviations from zero, yet $BF_{01}(A_b = 0) \approx 3.5$ (Fig. Vb), because the likelihood gain does not compensate for spreading the prior over $(-1, 1)$; unlike for A, the preference changes sign with the prior width (crossing at $W \approx 0.3$). Of the Bayesian results only the Bayes factor depends on the prior; the interval of A does not (Sec. V C). A negative A_b is plausible given the misidentified $B^{\pm} \to K^{\pm}\pi^+\pi^-$ and $\pi^{\pm}K^+K^-$ decays; A_b is kept free in the nominal model and fixing it to zero is one of the systematic variants (Sec. VI).

Model Adequacy: Goodness of Fit and Posterior Predictive Check: the model does not reach $\chi^2/\text{ndof} \approx 1$ (164.1/88), mainly because the signal tails are heavier than a Gaussian, but $A_{\rm raw}$ changes by only -0.0002 with a Crystal Ball signal.

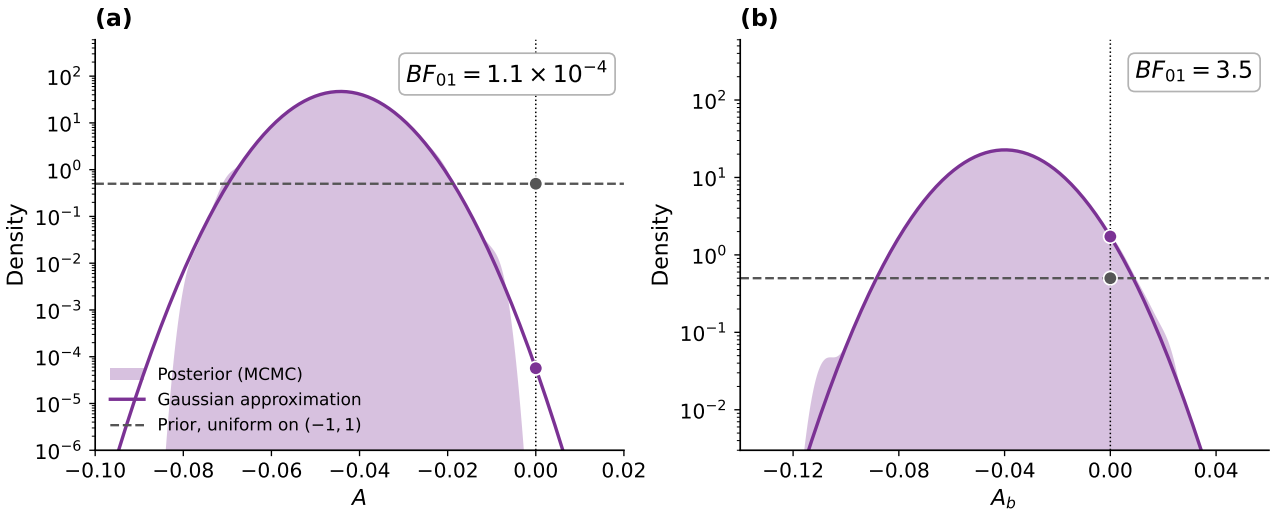


FIG. V. Savage-Dickey ratio for (a) A and (b) A_b: marginal posterior (histogram of the chain and Gaussian approximation) and prior density, logarithmic scale. The Bayes factor BF_{01} is the ratio of the two densities at zero (dots). For A the posterior density at zero is 10^{-4} times the prior one; for A_b, 2.3 standard deviations from zero, it is higher than the prior density, and the data favor $A_b = 0$.

E. Additional Results

Sampler. An own Metropolis-Hastings sampler reproduces the posterior of the ensemble sampler.

Intervals. Equal-tailed and highest-posterior-density 68% intervals coincide for A and A_b and differ for the skewed parameters; the profile likelihood is tabulated for combination with other results.

Priors. No change of the prior on the nuisance parameters or of its parametrization moves A by more than 0.0002, and the evidence by direct integration agrees in order of magnitude with the Savage-Dickey ratio.

Model adequacy. The model does not reach $\chi^2/\text{ndof} \approx 1$ (164.1/88), because the signal tails are heavier than a Gaussian, but A_{raw} changes by -0.0002 with a Crystal Ball signal; the posterior predictive check reproduces the asymmetry in slices of the mass ($p_B = 0.67$) and puts the difference of the χ^2 of the two charges at the edge of the replicas ($p_B = 0.08$).

Robustness and coverage. A_{raw} agrees between the magnet polarities (0.7σ) and the D^0 veto moves it by 2.3 times the expected spread; 200 toy-MC experiments give coverage 0.65 ± 0.03 (68%) and 0.93 ± 0.02 (95%) with pulls of width 1.04.

VI. SYSTEMATIC UNCERTAINTIES AND RESULT

In short: thirteen variants in four groups, fitted by both methods on the same candidates, give 0.0070 *(Bayesian) and* 0.0072 *(Frequentist), dominated by the particle-identification requirement.* The systematic study asks how much the result depends on choices of the analysis that the data do not fix. The Frequentist and Bayesian fits share one study, so that the two uncertainties are comparable: the same thirteen variants of the nominal choices, fitted by both methods on exactly the same candidates, and the same rule to combine them. The variants (listed in Table II) change one choice at a time and fall in four groups: the *window*, the *background* model, the *signal* model and the *preselection* cuts.

The shift $\Delta = A^{\text{var}}_{\text{raw}} - A^{\text{nom}}_{\text{raw}}$ of a variant is not necessarily a systematic effect: when a variant changes the sample (window, preselection) the two samples are nested and part of Δ is a statistical fluctuation of expected size $\sigma_B = \sqrt{|\sigma^2_{\text{var}} - \sigma^2_{\text{nom}}|}$ [13], and such a variant contributes only if $|\Delta| > \sigma_B$. A variant that the data reject ($\Delta\chi^2 > 25$ on the same data) is excluded: this removes the single exponential ($\Delta\chi^2 = 134$). The largest remaining $|\Delta|$ of each group is its uncertainty, since the variants of a group are alternative descriptions of the same choice, and the groups, being independent choices, are added in quadrature.

Figure VI shows the scan of each particle-identification cut, with the expected spread σ_B.

The results are in Table II. The two methods give the same shifts to within ∼0.0005 and the same group uncertainties, with totals of 0.0070 (Bayesian) and 0.0072 (Frequentist):

$$\text{Bayesian:}$$
$$A_{\text{raw}} = -0.0443 \pm 0.0085\,(\text{stat}) \pm 0.0070\,(\text{syst}),$$
$$\text{Frequentist:}$$
$$A_{\text{raw}} = -0.0441 \pm 0.0085\,(\text{stat}) \pm 0.0072\,(\text{syst}).$$

The preselection dominates, through the single variant `ProbK` > 0.6 $(+0.0058,\ 1.2\,\sigma_B)$. The scan of each cut

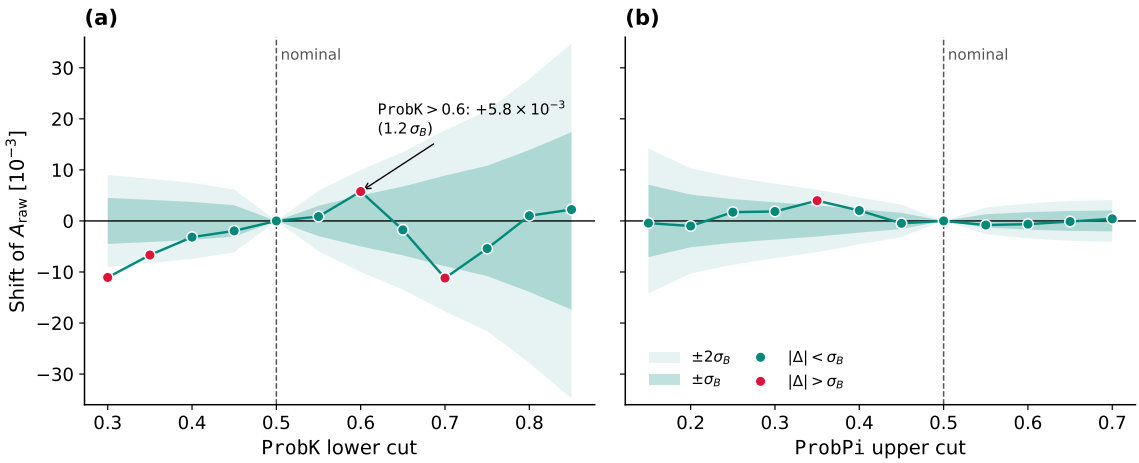


FIG. VI. Shift of A_{raw} (Frequentist fit) as a function of (a) the `ProbK` lower cut and (b) the `ProbPi` upper cut, in steps of 0.05, with the expected statistical spread of the nested sample ($\pm\sigma_B$ dark, $\pm 2\sigma_B$ light; zero at the nominal cut, dash-dotted line). On the tight side of `ProbK` the points scatter around zero without trend and the pion requirement is flat; loosening `ProbK` moves A_{raw} monotonically.

Group	Variant	Bayesian	Frequentist
Window	$[5050, 5600]$	-0.0003 (0.0009)	-0.0004 (0.0005)
	$[5150, 5600]$	-0.0001 (0.0007)	-0.0005 (0.0021)
	$[5100, 5450]$	$+0.0002$ (0.0009)	-0.0001 (0.0003)
	$[5100, 5700]$	-0.0014 (0.0010)	-0.0014 (0.0001)
	group uncertainty	*0.0014*	*0.0014*
Background	single exponential (rejected)	-0.0037	-0.0039
	shape shared between charges	-0.0037	-0.0040
	symmetric, $A_b = 0$	-0.0028	-0.0031
	group uncertainty	*0.0037*	*0.0040*
Signal	Crystal Ball	-0.0002	-0.0006
	μ, σ per charge	-0.0000	-0.0001
	group uncertainty	*0.0002*	*0.0006*
Preselection	`ProbK` > 0.4	-0.0032 (0.0038)	-0.0032 (0.0037)
	`ProbK` > 0.6	$+0.0058$ (0.0048)	$+0.0058$ (0.0050)
	`ProbPi` < 0.4	$+0.0022$ (0.0022)	$+0.0020$ (0.0023)
	`ProbPi` < 0.6	-0.0004 (0.0020)	-0.0006 (0.0017)
	group uncertainty	*0.0058*	*0.0058*
Total (quadrature)		0.0070	0.0072

TABLE II. Systematic study common to the Bayesian and Frequentist fits: shift Δ of A_{raw} for each variant and, in parentheses and for the variants that change the sample, the expected statistical spread σ_B of the shift. The largest contributing shift of each group is its uncertainty (italics); the groups are added in quadrature. The single exponential is rejected by the data and excluded. For the window $[5150, 5600]$ the Frequentist uncertainty is scaled from the number of events, since the Hesse matrix of that fit is not accurate.

in steps of 0.05 shows that this value is a fluctuation: on the tight side the shifts scatter around zero within $1.3\,\sigma_B$ without a trend, whereas loosening `ProbK` moves A_{raw} monotonically, as expected if misidentified decays not in the mass model enter the sample. The nominal requirement is therefore well placed and the preselection uncertainty conservative. The background group is second, driven by the charge dependence of the background, and the window third.

A. Additional Results

The σ_B rule. σ_B is the difference of two nearly equal uncertainties, so a variant at the threshold can be classified differently by the two methods without changing the result; the rule is verified by bootstrap (the spread of the shift between a sample and its tighter subsample is σ_B).

Stability scans. The Frequentist grid scans shift A_{raw} by at most 0.0048 over the window edges (55 of 80 fits valid), 6×10^{-5} over the signal region and 0.0126 over the grid of the two PID cuts, whose samples are smaller (mean uncertainty 0.0112).

VII. VALIDATION OF THE ESTIMATOR AND OF THE INTERVALS

In short: toy-MC experiments show an unbiased estimator at the minimum variance bound, intervals with nominal coverage and a test of $A = 0$ with the nominal size. Toy-MC experiments generated from the fitted model of Sec. III (Poisson yields, masses drawn from the model; 1000 at the fitted value $A_{\text{true}} = -0.0441$ and 300 at each of -0.1, -0.02, 0 and $+0.05$, 1–3% of the fits not valid and excluded) verify the estimator and the profile-likelihood ratio. The bias of $\hat{A}$ is at most 0.0008 (1.6 standard errors; -0.0005, -0.0001, -0.0008, $+0.0002$, -0.0004 for the five values), its spread equals the mean Hesse uncertainty within 6% and the pulls have width 0.94–1.05: the estimator is unbiased and reaches the minimum variance bound.

At the fitted value the coverage of $t(A_{\text{true}}) \leq 1$ is 0.696 ± 0.015 (68.3% level) and 0.954 ± 0.007 (95.4% level); the 68% coverage over the five values is 0.71, 0.70, 0.69, 0.69 and 0.61, the last, at $+0.05$ (0.614 ± 0.028), 2.5 standard deviations low, not unexpected among five values.

Inverting the Neyman construction of the same experiments at the observed $\hat{A}$ gives the 68% interval $[-0.0528, -0.0351]$, against the Minos interval $[-0.0526, -0.0356]$; Figure VII shows bias, pull width and power as a function of the injected asymmetry.

Under $A_{\rm true} = 0$ the test $q(0) > 3.84$ rejects in 0.051 ± 0.013 of the experiments, the nominal size, with $q(0)$ compatible with χ^2_1 (Kolmogorov-Smirnov $p = 0.79$); its power is 1.00 at the fitted value and 0.68 (0.27 at 3σ) at $A_{\rm true} = -0.02$, so with 19,731 candidates an asymmetry below ~ 0.02 is not resolved. The 5.2σ of the data is an extrapolation of the χ^2_1 tail, which the 293 experiments check only down to the 3σ level.

A. Additional Results

Asymptotic behavior. With the number of candidates the bias tends to zero, the spread scales as $N^{-1/2}$ and the distribution of $\hat{A}$ becomes Gaussian.

Goodness of fit and binning. The binned χ^2 of toy experiments follows its asymptotic distribution, so its p-values are accurate: the misfit of the data is 5.4 standard deviations in 10 MeV bins and 2.0 in 2.5 MeV bins, because it lives on the scale of the flanks of the peak.

VIII. LOCAL ASYMMETRY IN THE DALITZ PLANE

In short: the asymmetry is not uniform over the Dalitz plane ($\chi^2 = 80.8$ for 31 degrees of freedom) and is concentrated outside the $\phi(1020)$, in the region of the published analysis: -0.234 ± 0.025 against -0.239 ± 0.020. To test whether the asymmetry is uniform over phase space or concentrated in specific regions, each candidate was placed in the Dalitz plane using the two opposite-sign K^+K^- invariant masses $m^2(H1, H2)$ and $m^2(H1, H3)$ (Sec. II A), sorted into $m^2_{\rm lo}$ and $m^2_{\rm hi}$ to remove the $H2/H3$ labeling ambiguity. The plane $[0, 16] \times [0, 30]$ GeV^2 in $(m^2_{\rm lo}, m^2_{\rm hi})$ was divided into an 8×8 grid, fixed before looking at the asymmetries. In each bin with at least 20 candidates (32 of 64), the local asymmetry $A_i = (N_i^- - N_i^+)/N_i$, with binomial uncertainty $\sigma(A_i) = \sqrt{(1 - A_i^2)/N_i}$, was compared with the global counting asymmetry $A_{\rm glob} = -0.043$ through $S_i = (A_i - A_{\rm glob})/\sigma(A_i)$.

Figure VIII shows the occupancy of the bins (a), the local asymmetry (b) and its significance (c). Two of the 32 bins have $|S_i| > 3$, adjacent at $m^2_{\rm lo} < 4$ GeV^2 and $7.5 < m^2_{\rm hi} < 11.25$ GeV^2 ($S_i = -3.7$ and -3.0); about 0.09 such bins would be expected if the asymmetry were uniform, and two or more with probability 3×10^{-3}, a look-elsewhere number over any two of the 32 bins. The global homogeneity test, $\chi^2 = \sum_i S_i^2 = 80.8$ for 31 degrees of freedom ($p = 3 \times 10^{-6}$), confirms that the local asymmetry is not uniform; it also holds in the signal window alone, with the charm vetoes and for other grids.

The structure is the same with both magnet polarities: in the region of the two significant bins the counting asymmetry is -0.129 ± 0.023 (magnet down) and -0.126 ± 0.028 (magnet up), and the two maps agree bin by bin ($\chi^2 = 23.0$ for 29 bins, $p = 0.78$). The split tests the part of the detection asymmetry that changes sign with the polarity, not the different interaction of K^+ and K^- with the detector material.

Where is the asymmetry in terms of the decay dynamics? The Dalitz plane shows the $\phi(1020)$ band at the kinematic edge ($m^2_{\rm lo} \approx 1.04$ GeV^2) and the charm bands of the D^0, J/ψ and χ_{c0}. The negative asymmetry is not produced by the $\phi(1020)$: its band (1964 candidates) has $A_{\rm count} = -0.004 \pm 0.023$, while the significant bins without it have -0.145 ± 0.019. It sits just above the ϕ, where the published analysis located it, defining the region $1.2 < m^2_{\rm lo} < 2.0$ GeV^2, $m^2_{\rm hi} < 15$ GeV^2 with $A_{\rm raw} = -0.239 \pm 0.020$ [5]. The region is fixed by the paper, not chosen on our data, so the comparison is unbiased: the nominal model fitted to its 2146 candidates gives $A_{\rm raw} = -0.234 \pm 0.025$ (Bayesian) and -0.237 ± 0.025 (Frequentist, 9.4σ from zero), reproducing the published value.

Three caveats limit the interpretation: the binning is coarse and was not optimized, the asymmetries are counts without sWeight background subtraction, and local variations can be partly instrumental, since the $K^\pm$ interaction asymmetry depends on momenta that are correlated with the Dalitz position. In summary, the global asymmetry is not uniform over phase space, and the region identified by the published analysis carries it.

A. Additional Results

Null tests. Permuting the charges gives $\chi^2 = 31.5 \pm 8.4$ and never reaches 80.8 in 2000 permutations, and the phase-space simulation is consistent with no asymmetry (slightly optimistic p-values).

IX. DISCUSSION AND CONCLUSIONS

In short: the Frequentist and Bayesian analyses give the same $A_{\rm raw} = -0.044 \pm 0.009\,(stat) \pm 0.007\,(syst)$, 5.2 standard deviations from zero, with validated intervals; they differ in the meaning of the numbers and of the tests, not in their values. On this problem the Frequentist profile-likelihood fit and the Bayesian posterior are equivalent in value, uncertainty and coverage: with a common selection and mass model they give $A_{\rm raw} = -0.0441 \pm 0.0085 \pm 0.0072$ and $-0.0443 \pm 0.0085 \pm 0.0070$ (statistical, systematic), a nonzero asymmetry at 5.2 standard deviations, stable under the thirteen variants (the dominant systematic contribution is the particle-identification requirement) and concentrated in the Dalitz region of the

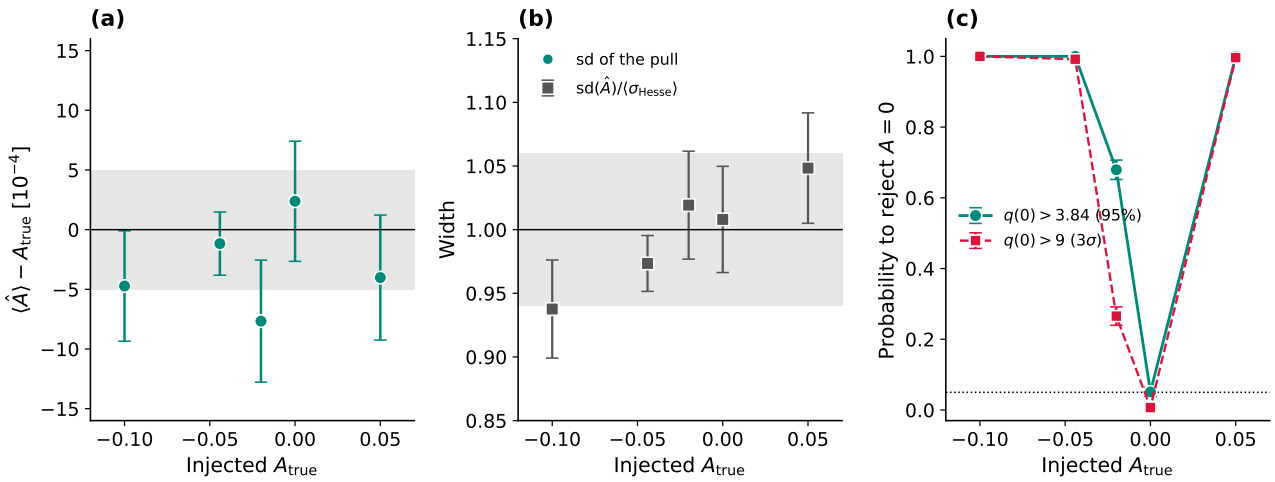


FIG. VII. Toy-MC experiments at five injected asymmetries. (a) Bias of $\hat{A}$, with its standard error. (b) Spread of the estimates divided by the mean Hesse uncertainty, and width of the pulls. (c) Probability to reject $A = 0$ in the test with $q(0) > 3.84$ (size 5%) and $q(0) > 9$ (3σ), as a function of the injected value: size at $A_{\rm true} = 0$, power elsewhere.

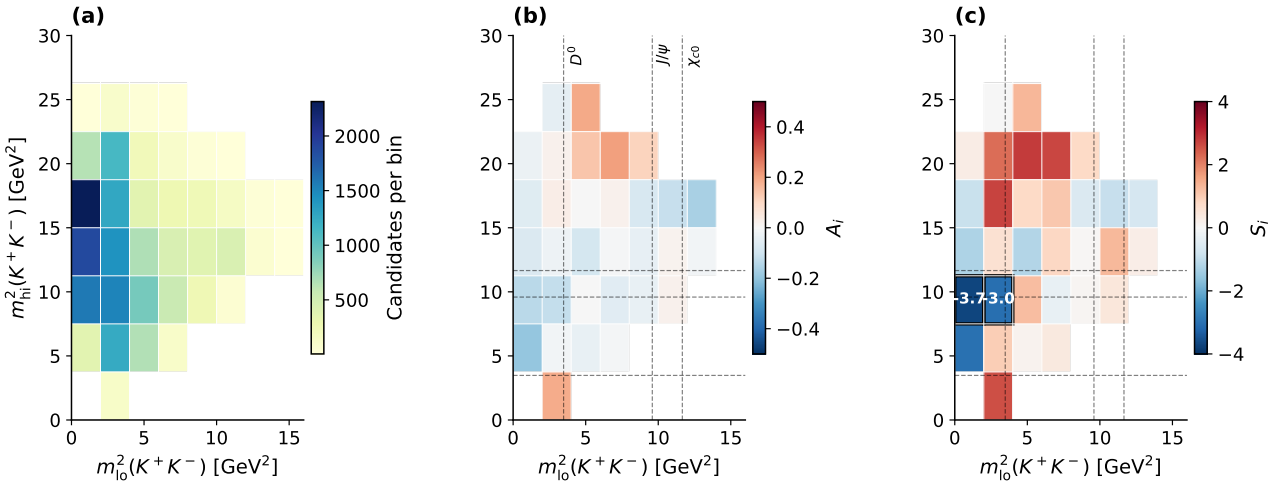


FIG. VIII. Fixed 8×8 Dalitz-plot binning, all candidates in the fit window: (a) number of candidates per bin, (b) local asymmetry A_i for bins with ≥ 20 candidates, and (c) local significance with respect to the global asymmetry, $(A_i - A_{\rm glob})/\sigma(A_i)$. The two bins with $|S_i| > 3$ are outlined. Dashed lines: the squared masses of the D^0, J/ψ and χ_{c0}, in either opposite-sign pair.

published analysis outside the $\phi(1020)$ (-0.234 ± 0.025 against -0.239 ± 0.020).

The numbers coincide, the statements do not. The credible interval states that A lies in $[-0.0528, -0.0358]$ with 68% probability given the data and the prior; the confidence interval $[-0.0526, -0.0356]$ is the set of values that the data do not reject, and its 68% is a property of the procedure that the bootstrap and the toy experiments verified. The Frequentist test of $A = 0$, $p = 2 \times 10^{-7}$, is independent of the prior; the Bayes factor, $BF_{10} \approx 9 \times 10^3$, depends on its width (Sec. V D), and for $A_b = -0.040 \pm 0.018$ a Frequentist test would call 2.3σ a hint while $BF_{01} \approx 3.5$ mildly favors $A_b = 0$. The Frequentist fit adds an unbiased estimator at the minimum variance bound; the Bayesian analysis adds sampling diagnostics and a posterior predictive check. Figure IX shows the profile likelihood and the marginal posterior of A on the same axis.

The published analysis [5] finds -0.060 ± 0.007 on a larger sample with a richer model; with the same D^0 veto the Bayesian fit gives -0.0496 ± 0.0088, within 0.93σ. The raw asymmetry is not the CP asymmetry, $A_{CP} \approx A_{\rm raw} - A_P - A_D$ (Eq. (2)): the 5.2σ here is the significance of a nonzero raw asymmetry, not of CP violation. Three limitations bound the result: no tested model reaches $\chi^2/{\rm ndof} \approx 1$, although $A_{\rm raw}$ does not depend on the signal shape; no production or detection correction is applied; and the local Dalitz-plane asymmetries are counts without background subtraction. A machine-learning selection leaves $A_{\rm raw}$ unchanged within its sta-

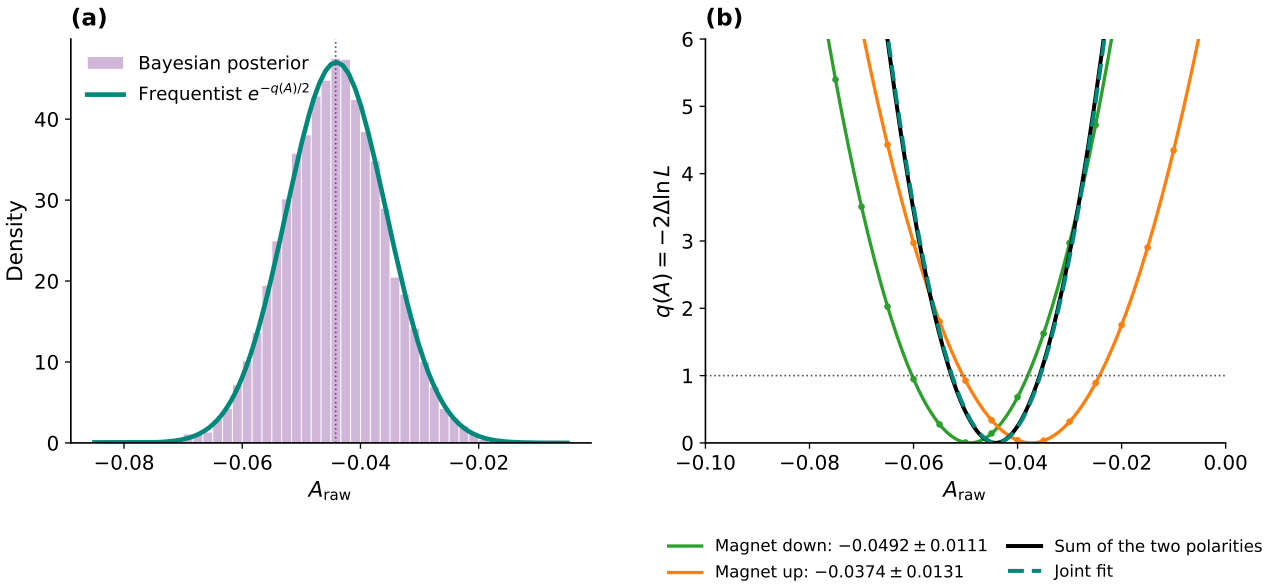


FIG. IX. (a) Frequentist profile likelihood, $\exp[-q(A)/2]$ normalized, and Bayesian marginal posterior of A. (b) Profile curves $q(A)$ of the two magnet polarities (parabolic fits to the points), their sum and the joint fit.

tistical spread, and the natural next step is a control-channel correction.

A. Additional Results

Numerical agreement. The estimates of the two fits differ by 0.0002, both uncertainties are 0.0085, and sideband subtraction gives -0.0503 ± 0.0083.

Profile, posterior and polarities. Profiling and marginalizing the eleven nuisance parameters give the same distribution of A, since A is almost uncorrelated with them; the two polarities agree and their combined likelihood reproduces the joint fit; the 68% and 95% regions in (A, A_b) of the profile (ΔNLL at 2.30 and 5.99) and of the highest posterior density coincide, with $A = 0$ outside both.

Tests and goodness of fit. The exact binomial test on all candidates gives 6.1σ. The same fit has $p = 0.03$ in 2.5 MeV bins and $p \approx 5 \times 10^{-6}$ in 10 MeV bins: the p-value of the binned χ^2 depends on the binning as much as on the model.

Low statistics. Credible and confidence intervals coincide in value and coverage from $\sim 2 \times 10^4$ down to 50 candidates.

Published analysis. It has $22{,}119 \pm 164$ signal candidates, a Cruijff signal and misidentified and partially reconstructed backgrounds; its corrections move -0.060 to $-0.043 \pm 0.009 \pm 0.003 \pm 0.007$ and cannot be assumed for another selection.

Machine learning: labels and models. A classifier separates the signal-enriched window from the sidebands (15,130 and 1940 candidates, 20 features, 60/20/20 split) and, by the Neyman-Pearson argument of CWoLa, is also the signal/background classifier; the tree ensembles and the neural network reach test AUC 0.857–0.864, 0.09 above the linear models and indistinguishable within 0.012, with gradient boosting as reference.

Machine learning: regularization and uncertainty. The linear model does not overfit, gradient boosting overfits beyond ~ 150 stages without losing test AUC, and class weights bias the probabilities, corrected by isotonic recalibration (Brier score $0.122 \to 0.062$); the mean per-candidate uncertainties are 0.178 (Random Forest), 0.095 (MC dropout) and 0.050 (deep ensemble).

Machine learning: Dalitz plane and selection. K-means with $K = 8$ finds two clusters beyond 3σ ($\chi^2 = 46.4$ for 7 degrees of freedom), the pattern of the fixed binning (80.8/31): negative at low m^2_{lo}, positive at high m^2_{hi}. A cut at 0.5 on the score keeps 92% of the signal and 36% of the background and gives -0.0451 ± 0.0087 (Bayesian) and -0.0447 ± 0.0086 (Frequentist), within the spread of a subsample, with counting too; the hyperparameter search gains at most 0.005 in AUC (winner's curse), and the selection cannot remove the misidentified background.

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